

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458185.6726 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.057
Transit depth (Rp/Rs)^2	2.67 +/- 1.87 [%]
Orbital Inclination (inc)	77.18 +/- 2.85
Airmass coefficient 1 (a1)	1.64 +/- 0.86
Airmass coefficient 2 (a2)	-0.27 +/- 0.1
Scatter in the residuals of the lightcurve fit is	50.37 %
Best Comparison Star	None
Optimal Aperture	3.29
Optimal Annulus	8.34
Transit Duration (day)	0.022 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.163 $\pm$ 0.057 vs 0.1594 (deviation 0.06 σ)
Duration fit vs published	0.022 d vs 0.0512 d (deviation 1.38 σ)
Mid-time O-C	-3.5 min
Depth SNR	2.7
Residual scatter	50.37 %

Light curve

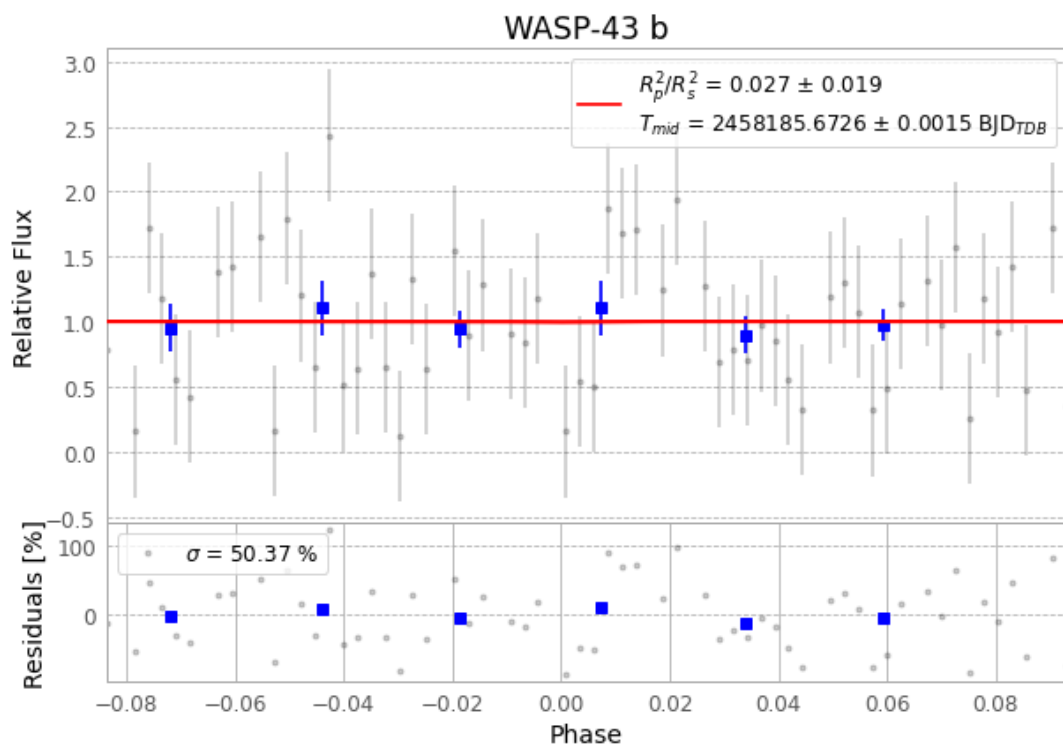


Figure 1 — FinalLightCurve_WASP-43 b_2018-03-07.png (SHA-256 33d9300b3e2bbe15...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [576, 416], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 80b6000fa3ce3ae0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

69 FITS frames (45 MB), WASP-43180308022113.FITS → WASP-43180308054812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180308.log (9b52c912c380...), FinalLightCurve_WASP-43 b_2018-03-07.png (33d9300b3e2b...), FinalLightCurve_WASP-43 b_2018-03-07.csv (b3094711b6f5...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 09:02Z.